5 Calculation of SARON Compound Rates

5.1 Formula

The SARON Compound Rates are calculated in arrears for pre-defined past time periods using the following formula.

$$SARON\ Compound\ Rate = \left[\prod_{i=1}^{bd}\left(1+\frac{r_i a_i}{360}\right)-1\right]\frac{360}{n}$$

bd	Number of trading days for an observation period from the start date (inclusive) to the end date (exclusive) date. For example, "bd" is equal to one (1) for an observation period from Monday to Tuesday
i	Index from one to bd
n	Number of calendar days of the observation period from the start date (inclusive) to the end date (exclusive). For example "n" is equal to one (1) for an observation period from Monday to Tuesday
r_i	SARON on business day i
a_i	Number of calendar days for which SARON ri applies

Alternatively, the SARON Index (SAION), which also belongs to the family of Swiss Reference Rates, can be used to calculate the SARON Compound Rates. The SARON Index reflects the performance of SARON through daily compounding. Further details on the methodology and formula of the SARON Index are described in section 4.

$$SARON\ Compound\ Rate = \left(\frac{SARON\ Index_E}{SARON\ Index_S}-1\right)\frac{360}{n}$$

n	Number of calendar days for an observation period from the start date S (inclusive) to the end date E (exclusive)
	SARON $Index_S$ and SARON $Index_E$ - SARON Index value at the start date S and end date E

The advantage of using the SARON Index is that only two data points are needed to calculate the compounded SARON value for a given tenor, while the standard formula requires daily data of the SARON value. Both formulas can be used to calculate a compounded SARON for any combination of trading days.

Since the SARON Index reflects the same arithmetic as a compounded SARON and the SARON Compound Rates, calculations using the SARON Index with the same start and end dates should effectively produce equivalent results. However, because the SARON Index is rounded, its values do not have the same precision as the compounded SARON. Therefore, minor differences may occasionally occur at the fourth decimal place.

The SARON Compound Rates are calculated to four decimal places and rounded with standard commercial practice.

5.2 Definition of Start and End Dates for SARON Compound Rates

The SARON Compound Rates are provided for pre-defined tenors. The end date for each period is the current business day on which the underlying rate, SARON, is calculated. The start date for the daily SARON Compound Rates is the business day that is the corresponding number of weeks/months prior to the end date. The determination of the start date is aligned with the CHF money market calendar.[10] If a determined start date falls on a non-business day like a weekend or a currency holiday, the start date will be adjusted.

In the CHF money market calendar, the end date is set in advance based on the start date and the non-business days are adjusted according to the Modified Following Business Day Convention. There are three scenarios:

- If the start date falls on the last business day of a month, the end date must also fall on the last business day of a month.
- If the end date falls on a non-business day, the time period is extended by moving it to the next business day unless it falls in a new month.
- If the postponement of the date causes the end date to fall in a new month, the period is shortened by moving it to the previous business day.

Note: For SARON Compound Rates with a tenor of less than one month (e.g. weekly), the scenarios do not apply, as the month-end restrictions are obsolete in such a case.

The SARON Compound Rates are calculated retrospectively, therefore the Modified Following Business Day Convention cannot be applied directly. In order to align as closely as possible with the CHF money market calendar and the Business Day Convention, the start date is determined as follows:

- If the date is unique according to the CHF money market calendar, it will be used as the start date.
- If the end date falls on the last business day of a month, the start date must also be the last business day of a month.
- For each end date with several possible start dates according to the CHF money market calendar, the following applies (unless the end date is the last business day of a month:
  - In case of an uneven number of possible start dates, the middle date will be chosen as the start date
  - In case of an even number of possible start dates, the earlier of the two middle dates will be chosen
- If the originally calculated start date falls on a non-business day or non-existent date (e.g. 30th of February), the business day preceding the calculated start date will be used as the start date, unless this new start date would fall in a different month. In this case, the following business day will be used as the start date and not the previous business day.

In general, SARON Compound Rates with a tenor of less than one month (e.g. weekly) simplify the determination of the start- and end dates, since the month-end restrictions are omitted. However, the other conventions of the money market calendar must be maintained.

5.3 Definition of Start and End Dates of the SARON IMM Compound Rates

SIX provides the SARON IMM Compound Rates for pre-defined tenors. According to the IMM (International Money Market Calendar), the end date of these tenors is the 3rd Wednesday of a month and is by definition always a business day. The start date is the 3rd Wednesday of a month and is the corresponding number of months before the end date.

[10] https://www.six-group.com/en/market-data/news-tools/trading-currency-holiday-calendar.html#/currencyCalendar

5.4 Examples for the Definition of Start Dates

The following table provides examples of how to define the end and start dates for SARON Compound Rates.

Time Period	End Date	Start Date	Comment
1 month	30.04.2018	29.03.2018	The end date falls on the last business day of the month. The start date is moved to the last business day of a month.
1 month	15.06.2018	15.05.2018	Unique allocation according to the money market calendar.
1 month	08.10.2018	06.09.2018 or 07.09.2018	Two possible start dates according to the money market calendar, leading to the end date 08.10.2018. The earlier date 06.09.2018 will be selected.
1 month	23.04.2018	21.03.2018 or 22.03.2018 or 23.03.2018	Three possible start dates according to the money market calendar, leading to the end date 23.04.2018. The middle date 22.03.2018 will be selected.
1 month	10.12.2019	08.11.2019	The previous business day is used since 10.11.2019 is not a non-business day.

The compounding of historical interest rates ("compounding in arrears") and the money market calendar result in the same dates for the end-of-month periods. However, there might be differences in the day count due to currency holidays such as Easter and Christmas. The impact on calculations is small, especially for longer tenors.

5.5 Calculation Example SARON 1 Month Compound Rate

end_date	08.10.2018
start_date (two possible start dates according to the money market calendar, leading to the end date 08.10.2018. The earlier date 06.09.2018 will be selected)	06.09.2018
day_count	32
SARON Compound Rate: (product (multiplier) -1) *360 / day_count)	-0.7451
SARON Compound Rate: (SARON Index (end) / SARON Index(start) -1)* (360/ day_count)	-0.7451

Date	SARON	multiplier (1+ SARON x day_count / 360)	day_count	SARON Index
06.09.2018	-0.739773	0.999979	1	11048.90141
07.09.2018	-0.737137	0.999938572	3	
10.09.2018	-0.73405	0.99997961	1	
11.09.2018	-0.742549	0.999979374	1	
12.09.2018	-0.744533	0.999979319	1	
13.09.2018	-0.739139	0.999979468	1	
14.09.2018	-0.734535	0.999938789	3	
17.09.2018	-0.732281	0.999979659	1	
18.09.2018	-0.739414	0.999979461	1	
19.09.2018	-0.741015	0.999979416	1	
20.09.2018	-0.740611	0.999979427	1	
21.09.2018	-0.743656	0.999938029	3	

Date	SARON	multiplier (1+ SARON x day_count / 360)	day_count	SARON Index
24.09.2018	-0.736047	0.999979554	1	
25.09.2018	-0.745040	0.999979304	1	
26.09.2018	-0.760342	0.999978879	1	
27.09.2018	-0.753971	0.999979056	1	
28.09.2018	-0.785767	0.999934519	3	
01.10.2018	-0.738704	0.99997948	1	
02.10.2018	-0.734949	0.999979585	1	
03.10.2018	-0.743903	0.999979336	1	
04.10.2018	-0.742927	0.999979363	1	
05.10.2018	-0.746194	0.999937817	3	
08.10.2018				11041.58344

5.6 Calculation Interval and Publication Times

SARON Compound Rates and Indices are published after close of trading. They are available on the Index Data Center webpage around 18.50 CET. At 19:00 CET, the market data is distributed via SIX Exfeed.[11]

[11] https://indexdata.six-group.com/

6 Additional Information

To assist customers in determining a compounded SARON outside of the standard 1, 3 and 6 months periods, SIX provides the "SARON Compound calculation matrix". This file contains compounded SARON values for all date combinations, including weekends and non-business days of the last 12 months. In addition, a web-based calculator for a compounded SARON is available for ad-hoc calculations.[12]

6.1 Formula for a Compounded SARON on Non-Business Days

For cases where a compounded SARON is required for a non-business day (e.g. starting or ending on a weekend), the National Working Group on Swiss Franc Reference Rates has developed the following approximation to cover non-business days.

$$compounded\ SARON = \left[\prod_{i=1}^{bd}\left(1+\frac{r_i a_i}{360}\right)-1\right]\frac{360}{n}$$

bd Number of business days for an observation period from the start date (inclusive) to the end date (exclusive), except if the start date is not a business day, then bd is increased by one. For example, bd is equal to one (1) for an observation period from Monday to Tuesday, two (2) for an observation period from Sunday to Tuesday and one (1) for an observation period from Friday to Sunday.

i Index from one (1) to bd

n Number of calendar days of the observation period from the start date (inclusive) the end date (inclusive). For example, n is equal to one (1) for an observation period from Monday to Tuesday.

r_i SARON on business day i. If the start date is not a trading day, the SARON from the previous business day is used.

a_i Number of calendar days for which SARON r_i applies. If the observation period ends on a Sunday, a_i normally includes two days (2). If the observation period starts on a Sunday, a_i is equal to one (1) and the SARON of the previous business day is used.

[12] https://www.six-group.com/en/market-data/indices/switzerland/saron/calculator.html